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Dormancy-Associated MADS-Box (*DAM*) and the Abscisic Acid Pathway Regulate Pear Endodormancy Through a Feedback Mechanism

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In the pear 'Kosui' (*Pyrus pyrifolia* Nakai), the *dormancy-associated MADS-box* (*PpDAM1* = *PpMADS13-1*) gene has been reported to play an essential role in bud endodormancy. Here, we found that PpDAM1 up-regulated expression of *9-cis-epoxycarotenoid dioxygenase* (*PpNCED3*), which is a rate-limiting gene for ABA biosynthesis. Transient assays with a dual luciferase reporter system (LUC assay) and electrophoretic mobility shift assay (EMSA) showed that PpDAM1 activated *PpNCED3* expression by binding to the CArG motif in the *PpNCED3* promoter. *PpNCED3* expression was increased toward endodormancy release in lateral flower buds of 'Kosui', which is consistent with the induced levels of ABA, its catabolism (*ABA 8′-hydroxylase*) and signaling genes (*type 2C protein phosphatase* genes and *SNF1-related protein kinase 2* genes). In addition, we found that an ABA response element (ABRE)-binding transcription factor, PpAREB1, exhibiting high expression concomitant with endodormancy release, bound to three ABRE motifs in the promoter region of *PpDAM1* and negatively regulated its activity. Taken together, our results suggested a feedback regulation between *PpDAM1* and the ABA metabolism and signaling pathway during endodormancy of pear. This first evidence of an interaction between a *DAM* and ABA biosynthesis in vitro will provide further insights into bud endodormancy regulatory mechanisms of deciduous trees including pear.

Keywords: ABRE-binding protein transcription factor • Abscisic acid • *9-cis-epoxycarotenoid dioxygenase* • *Dormancy-associated MADS-box* • Pear (*Pyrus pyrifolia* Nakai).

Abbreviations: *ABA8′OH*, *ABA 8′-hydroxylase*; ABRE, ABA response element; AREB/ABF, ABRE-binding protein/ABRE-binding factor; CBF, C-repeat binding factor; *DAM*, *dormancy-associated MADS-box*; DVI, development index; EMSA, electrophoretic mobility shift assay; FLUC, firefly luciferase; *FT*, *FLOWERING LOCUS T*; HC, hydrogen cyanamide; LUC, luciferase; *NCED*, *9-cis-epoxycarotenoid dioxygenase*; ORF, open reading frame; *PP2C*, *type 2C protein phosphatase*; qRT–PCR, quantitative real-time reverse transcription–PCR; RLUC, Renilla luciferase; *SnRK2*, *SNF1-related protein kinase 2*; *SVP*, *SHORT VEGETATIVE PHASE*.

Footnotes: The nucleotide sequences of genes identified in this paper have been submitted to the DNA Data Bank of Japan (DDBJ): *PpNCED3* (LC155801), *PpABA8′OH* (LC155802), *PpPP2C-1* (LC155803), *PpPP2C-2* (LC155804), *PpPP2C-3* (LC155805), *PpSnRK2-1* (LC155806), *PpSnRK2-2* (LC155807), *PpSnRK2-3* (LC155808) and *PpAREB1* (LC155809).

Introduction

Winter is a challenge for the survival of deciduous fruit trees such as pears and apples. Deciduous fruit trees are exposed not only to low temperatures but also to drought stresses in the winter season. To survive such serious environmental conditions, deciduous fruit trees cease their growth in late autumn to early winter, shed their leaves and enter the endodormancy stage, which is regulated by internal physiological factors (Lang 1987, Crabbe and Barnola 1996, Arora et al. 2003, Rohde et al. 2007). For completion of endodormancy, some extent of chilling is required, and this amount is genetically determined based on the cultivars and strains (Lang 1987, Atkinson et al. 2013); for example, it is known that one of the major Japanese pear cultivars, 'Kosui', requires about 750 h chilling accumulation at 0–6°C to complete the endodormancy stage (Sugiura et al. 2011). After endodormancy, fruit trees enter the ecodormancy stage, in which sprouting of buds is still depressed due to the unsuitable environmental conditions (Lang 1987). Subsequently buds enter the sprouting stage after meeting a specific heat unit in the spring season.

Recently, insufficient chilling accumulation in the winter season due to global warming, leading to the inability to complete endodormancy, has become a big challenge (e.g. Kuroki et al. 2013). Development of a new technique for sustained stable production of deciduous fruits could be a pressing issue. Therefore, it is necessary to uncover the molecular mechanism underlying the establishment and release of endodormancy. Several studies have been carried out to elucidate the molecular

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mechanism of dormancy in deciduous trees, most focusing on poplar (Rohde et al. 2007, Cooke et al. 2012). Noticeably, *CONSTANS* (*CO*), *FLOWERING LOCUS T* (*FT*), *CENTRORADIALIS* (*CEN*) and *TERMINAL FLOWER1* (*TFL1*) have been reported to play important roles in dormancy establishment and release in *Populus trichocarpa* (Böhlenius et al. 2006) and *Populus* spp. (Mohamed et al. 2010, Hsu et al. 2011). Ibáñez et al. (2010) showed that the activities of circadian-related genes were involved in the entry to and exit from seasonal dormancy in *Populus tremula* × *Populus tremuloides*. Another interesting insight comes from the discovery of the peach mutant, *evg*, which shows a continuous growth habitat without cessation of shoot growth. This peach mutant lacks four MADS-box transcription factor genes, designated as *dormancy-associated MADS-box* (*DAM*) *1–4* genes, and this deletion also causes the loss of transcriptional activity in the other two MADS-box genes *DAM5* (*EVG.14*) and *DAM6* (*EVG.15*) (Bielenberg et al. 2008). In peach, the expression of *DAM5* (=*EVG.14*) and *DAM6* (= *EVG.15*) was high during endodormancy and low with the release of endodormancy (Jiménez et al. 2010, Yamane et al. 2011). Several studies demonstrated that the expression of homologs of these peach *DAM5* and *DAM6* genes in other plants was up-regulated towards endodormancy establishment, and showed a sharp decrease that is concomitant with endodormancy release. A transgenic study on overexpressing Japanese apricot (*Prunus mume*) *DAM* in poplar showed that the Japanese apricot *DAM6* gene possessed a strong inhibitory effect on growth even under favorable conditions (Sasaki et al. 2011). Horvath et al. (2008) found that expression of leafy spurge *DAM1* and *DAM2* was associated with endodormancy induction in leafy spurge. In pear, the expression level of *PpMADS13-1*, which is comparable with peach *DAM6* (Bielenberg et al. 2008), was induced towards endodormancy establishment and then decreased concomitant with endodormancy release (Saito et al. 2013). All these results suggest that *DAM* genes including the pear *PpMADS13-1* gene, which is named *PpDAM1* in this study, could play a crucial role in endodormancy phase transition. Therefore, it could be interesting to identify the genes downstream of *PpDAM1* for global understanding of the signal network during endodormancy phase transition.

The *DAM* gene belongs to the flower regulator clade of MIKC-type MADS-box transcription factor genes that includes *SHORT VEGETATIVE PHASE* (*SVP*) and *AGAMOUS-LIKE 24* (*AGL24*) (Jiménez et al. 2009). *SVP*, an Arabidopsis homolog of *DAM*, directly inhibited *FT* expression through the binding to the CArG motif of the *FT* promoter (Lee et al. 2007). The association of DAM with *FT* has been investigated in leafy spurge (Horvath et al. 2008) and pear (Saito et al. 2015, Niu et al. 2016). However, the detailed mechanism of this association during endodormancy is still unclear. Other than *FT* genes, to date no genes have been known to act downstream of *DAM*. On the other hand, we found that there is a CArG box in the putative promoter region of a *9-cis-epoxycarotenoid dioxygenase* gene (*PpNCED3*), a key regulatory gene for ABA biosynthesis (Seo and Koshiba 2002, Xiong and Zhu 2003). Since the role of ABA in bud dormancy has become apparent in the recent years, we hypothesize that *PpNCED3* may act as a downstream gene of *PpDAM1* to regulate endodormancy in pear.

In general, ABA plays important roles in biotic and abiotic stress responses, controlling a wide range of essential physiological processes. Most of the genes involved in ABA metabolism and signaling pathways have been identified in plants. ABA is synthesized from C_{40} carotenoid precursors via several enzymatic steps in which *NCED* is thought to be the key regulator (Seo and Koshiba 2002, Schwartz et al. 2003). The ABA level is regulated with the metabolization of ABA into an inactive product through a hydroxylation reaction catalyzed by ABA 8′-hydroxylase (ABA8′OH) (Schwartz et al. 2003, Xiong and Zhu 2003). The ABA signaling mechanism exhibits differential modification in the presence or absence of ABA. In the absence of ABA, type 2C protein phosphatases (PP2Cs) inhibit the activities of SNF1-related protein kinase 2s (SnRK2s) by dephosphorylating their kinase activation loop (Finkelstein 2013, Ng et al. 2014). In the presence of ABA, ABA receptors form a complex with PP2Cs in which the phosphatase activity of PP2Cs is inhibited, facilitating activation of SnRK2s (Finkelstein 2013, Ng et al. 2014). Subsequently, SnRK2s activate ABRE-binding protein/ABRE-binding factor (AREB/ABF) transcription factors, which then initiate transcription at ABA-responsive promoter motifs (ABREs) in the promoter regions of their target genes (Ng et al. 2014).

The involvement of ABA in seed dormancy has been well documented (Finkelstein 2013). Although both bud and seed dormancy are regulated by ABA, studies on the role of ABA in bud dormancy are still lagging behind compared with those on seed dormancy. In poplar, ABA biosynthetic genes (*ABA1*, *ABA2* and *NCED*) and ABA signaling genes (*PP2C* and *AREB*/*ABF*) were up-regulated after growth cessation during bud dormancy along with an increase in ABA levels (Rohde et al. 2002, Arora et al. 2003, Rohde et al. 2007, Ruttink et al. 2007). An increase in ABA levels, together with expression of *NCED*, was observed prior to the onset of bud dormancy; subsequently, a decrease in the levels of ABA content and *NCED* expression and an up-regulated expression level of the ABA catabolism gene, *ABA8′OH*, were observed after buds received enough chilling accumulation in grapes (Or et al. 2000, Zheng et al. 2015) and peaches (Wang et al. 2016). Decreased levels of ABA and ABA biosynthetic gene expression and increased levels of ABA catabolism genes were also found during the transition from endodormancy to ecodormancy in pear buds (Tamura et al. 1992, Bai et al. 2013). In addition, ABRE motifs are over-represented in the upstream sequences of dormancy-regulated genes in poplar, suggesting that the dormancy-associated changes in expression of these genes are regulated by the ABA signaling pathway (Howe et al. 2015).

Because of the economic importance of pear (*Pyrus pyrifolia* Nakai), which is ranked third in terms of fruit production in Japan, the bud dormancy process of pear has become an important concern under global warning condition and it has also attracted much interest of researchers and breeders. The dormancy status of flower buds is estimated using development index (DVI) models, consisting of two models, DVI_1 and DVI_2 (Oya 2006, Sugiura et al. 2011). Based on DVI values (Oya 2006, Sugiura et al. 2011), the flower bud status of pears is divided into three stages during the dormancy process: (i) $DVI_1 = 0–1.0$,

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endodormancy; (ii) $DVI_1 = 1.0–2.2$, transition stage from endodormancy to ecodormancy; and (iii) $DVI_2 = 0–1.0$, ecodormancy to flowering. Using these models, in this study, the transcriptional regulation of *PpDAM1* and ABA metabolism- and signaling-related genes was studied during seasonal endodormancy in lateral flower buds. We also investigated the transcriptional regulation of these genes with and without treatment with hydrogen cyanamide (HC), a well-known bud dormancy-releasing compound. We confirmed the association of PpDAM1 protein and the *PpNCED* promoter using promoter analysis in transient assays with the dual luciferase reporter system (LUC assay) and electrophoretic mobility shift assay (EMSA). In addition, several ABRE-binding motifs of AREB/ABF transcription factors were found in the upstream sequences of *PpDAM1*, leading to our investigation of the association of AREB/ABF protein and *PpDAM1*. The results suggested the possibility of a feedback regulation of *PpDAM1* in PpDAM1–PpNCED–ABA–PpAREB/ABF cycles in lateral flower buds during endodormancy.

Results

Changes in mRNA and protein levels of *PpDAM1* in pear lateral flower buds from endodormancy to the transition stage to ecodormancy

First, to select an internal control for gene expression analysis, we designed primers targeted to six frequently used reference genes for quantitative real-time PCR in pear: β-tubulin (*bTUB*), *Histone H3* (*HIS*), *Glyceraldehyde-3-phosphate dehydrogenase* (*GAP*), *Actin* (*ACT*), *Annexin* (*ANX*) and *SAND*. The expression stability analysis evaluated by NormFinder software packages (Supplementary Table S1) revealed that *SAND* was expressed quite stably in all sample groups (Andersen et al. 2004). Taken together with our previous report in which *SAND* is ranked as one of the best genes (Imai et al. 2014), we chose *SAND* as an internal control for gene expression analysis in this study.

The expression of *PpDAM1* was investigated in 'Kosui' lateral flower buds every 2 weeks from early autumn to late winter in the growing seasons of 2015–2016. The mRNA level of *PpDAM1* increased steadily from $DVI_1 = 0.03$, reached the highest level at $DVI_1 = 0.41$, then decreased until $DVI_1 = 2.02$ (**Fig. 1A**). To check the level of PpDAM1 protein using Western blotting, an antibody recognizing the peptide sequence of PpDAM1 (CSDTLSLKLGLP) was used. However, there are two other isoforms of DAM (PpDAM2 = PpMADS13-2 and PpDAM3 = PpMADS13-3) in pear that share very high sequence identity with PpDAM1 (Supplementary Fig. S1) (Saito et al. 2013). Therefore, the antibody recognizing the peptide sequence of PpDAM1 can also detect PpDAM2 and PpDAM3 proteins. The predicted molecular mass of the PpDAM1 protein is 25.09 kDa calculated by the ExPASy pI/Mw tool (http://web.expasy.org/compute_pi/), while PpDAM2 and PpDAM3 contain seven amino acids more than PpDAM1 and have predicted mol. wts. of 26.39 and 26.17 kDa, respectively (Supplementary Fig. S1). As seen in **Fig. 1B**, bands representing all the isoforms were detected by the anti-PpDAM1 antibody. Both upper and lower bands showed increases in expression from $DVI_1 = 0.09$ to $DVI_1 = 0.41$, and then a decrease at $DVI_1 = 1.37$. It seems likely that the upper bands were from PpDAM2 and PpDAM3 proteins, which have similar amino acid lengths as well as very similar predicted molecular weights and could not be discriminated on a 14% SDS–polyacrylamide gel. The lower bands probably represented PpDAM1 protein. However, the genomic structure of PpDAM1 consisted of eight exons and seven introns flanked with start and

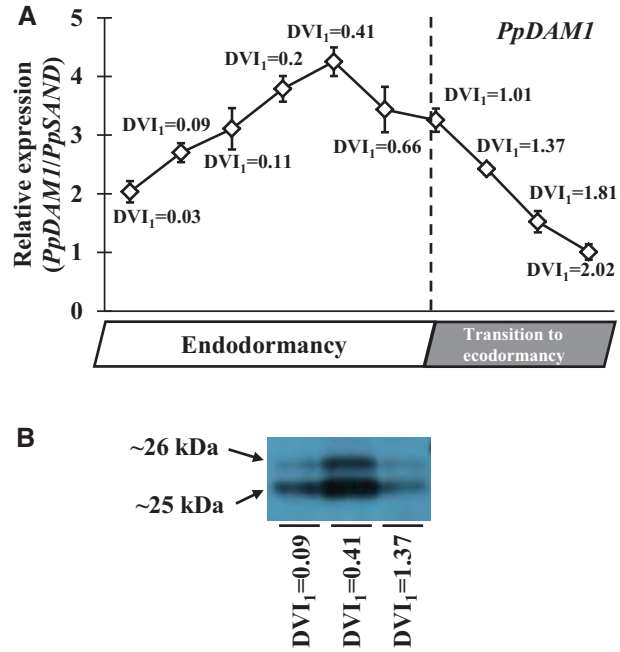


Fig. 1 Real-time quantitative PCR (A) and Western blot (B) analyses of *PpDAM1* expression during seasonal endodormancy in 'Kosui' lateral flower buds. Expression values were normalized to the value of the *PpSAND* gene. Each value represents the mean ± SE of three biological replicates, each with three technical replicates. Sampling dates are shown in **Table 1**. The dashed line indicates the date corresponding to endodormancy release.

Table 1 Sampling dates of 'Kosui' flower buds in the 2014–2015 and 2015–2016 seasons

Sampling date	
2014–2015	**2015–2016**
October 16 (0.01)	October 26 (0.03)
November 17 (0.19)	November 7 (0.09)
December 4 (0.45)	November 17 (0.11)
December 16 (0.8)	November 27 (0.2)
January 3 (1.3)	December 8 (0.41)
January 26 (1.98)	December 22 (0.66)
	January 5 (1.01)
	January 19 (1.37)
	February 4 (1.81)
	February 17 (2.02)

The numbers in parentheses are the DVI_1 values on each sampling date.

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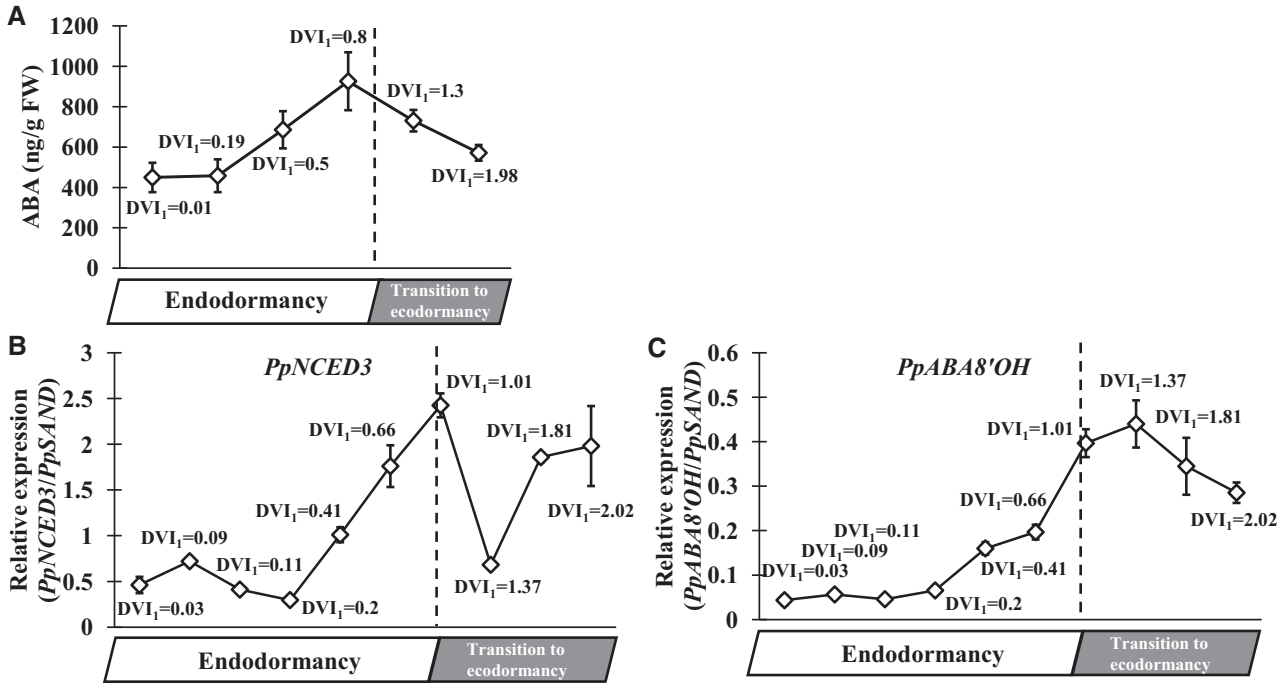


Fig. 2 ABA content (A), and expression levels of *PpNCED3* (B) and *PpABA8′OH* (C) during seasonal endodormancy in 'Kosui' lateral flower buds. Expression values were normalized to the value of the *PpSAND* gene. Each value represents the mean ± SE of three biological replicates, each with three technical replicates. Sampling dates are shown in **Table 1**. Dashed lines indicate the date corresponding to endodormancy release.

stop codons (Saito et al. 2013); therefore, the bands observed in **Fig. 1B** may also be from protein isoforms generated from alternative splicing of *PpDAM1*. Indeed, we recognized the possibility of alternative splicing of *PpDAM1* during pear endodormancy when we analyzed RNA sequencing data for preparation of our previous paper (Bai et al. 2013). Further extensive studies such as protein sequencing are needed to clarify the specificity of bands in **Fig. 1B**.

Isolation of *PpNCED3*, *PpABA8′OH*, *PpPP2Cs* and *PpSnRK2s* from pear

As mentioned in the Introduction, *PpNCED3* was considered as one of the downstream genes of PpDAM1 protein. The full-length open reading frame (ORF) of *PpNCED3* was then isolated in 'Kosui' using the *P. pyrifolia* transcriptome database (Liu et al. 2012, Bai et al. 2013). The ORF of *PpNCED3* was composed of 1,842 bp encoding a protein of 613 amino acids. Sequence analysis revealed that *PpNCED3* shares high sequence identity with *NCED* genes from other plants and *PpNCED3* is closer to *AtNCED3* (AT3G14440.1) compared with all other *NCED* genes from *Arabidopsis thaliana* (Supplementary Fig. S2). Using the information of *P. pyrifolia* transcriptome databases (Liu et al. 2012, Bai et al. 2013), we also screened and selected *PpABA8′OH*, *PpPP2C-1*, *PpPP2C-2*, *PpPP2C-3*, *PpSnRK2-1*, *PpSnRK2-2* and *PpSnRK2-3* for further analyses based on (i) high identities with other orthologous genes involved in ABA metabolism and signaling pathways and (ii) significant changes in expression levels during endodormancy.

ABA content and expression levels of ABA metabolism- and its signaling-related genes in pear lateral flower buds from endodormancy to the transition stage to ecodormancy

ABA content remained constant at the beginning of endodormancy, then increased at $DVI_1 = 0.5$ and reached its highest level at $DVI_1 = 0.8$, then gradually decreased until $DVI_1 = 1.98$ (**Fig. 2A**). The accumulation pattern of ABA relatively correlated with the expression patterns of its biosynthetic and catabolic genes. Specifically, *PpNCED3* and *PpABA8′OH* expression levels were low from $DVI_1 = 0.03$ to $DVI_1 = 0.2$, dramatically increased and peaked at $DVI_1 = 1.01$ (*PpNCED3*) or $DVI_1 = 1.37$ (*PpABA8′OH*), and then declined to lower levels during transition stage to the ecodormancy (**Fig. 2B, C**). Similar patterns in expression levels were observed in *PpPP2C-2*, *PpPP2C-3* and *PpSnRK2-2*, with low transcription levels at the beginning of endodormancy to mid endodormancy, then up-regulated expression from mid endodormancy to endodormancy release (**Fig. 3B, C, E**). Although *PpPP2C-1*, *PpSnRK2-1* and *PpSnRK2-3* also revealed increased expression toward endodormancy release, they showed high expression at the early stage of endodormancy (**Fig. 3A, D, F**).

Effects of HC treatment on expression levels of *PpDAM1* and ABA metabolism- and its signaling-related genes in pear lateral flower buds

In mid winter, 'Kosui' lateral flower buds of the potted trees were treated with HC and were transferred into a warm greenhouse.

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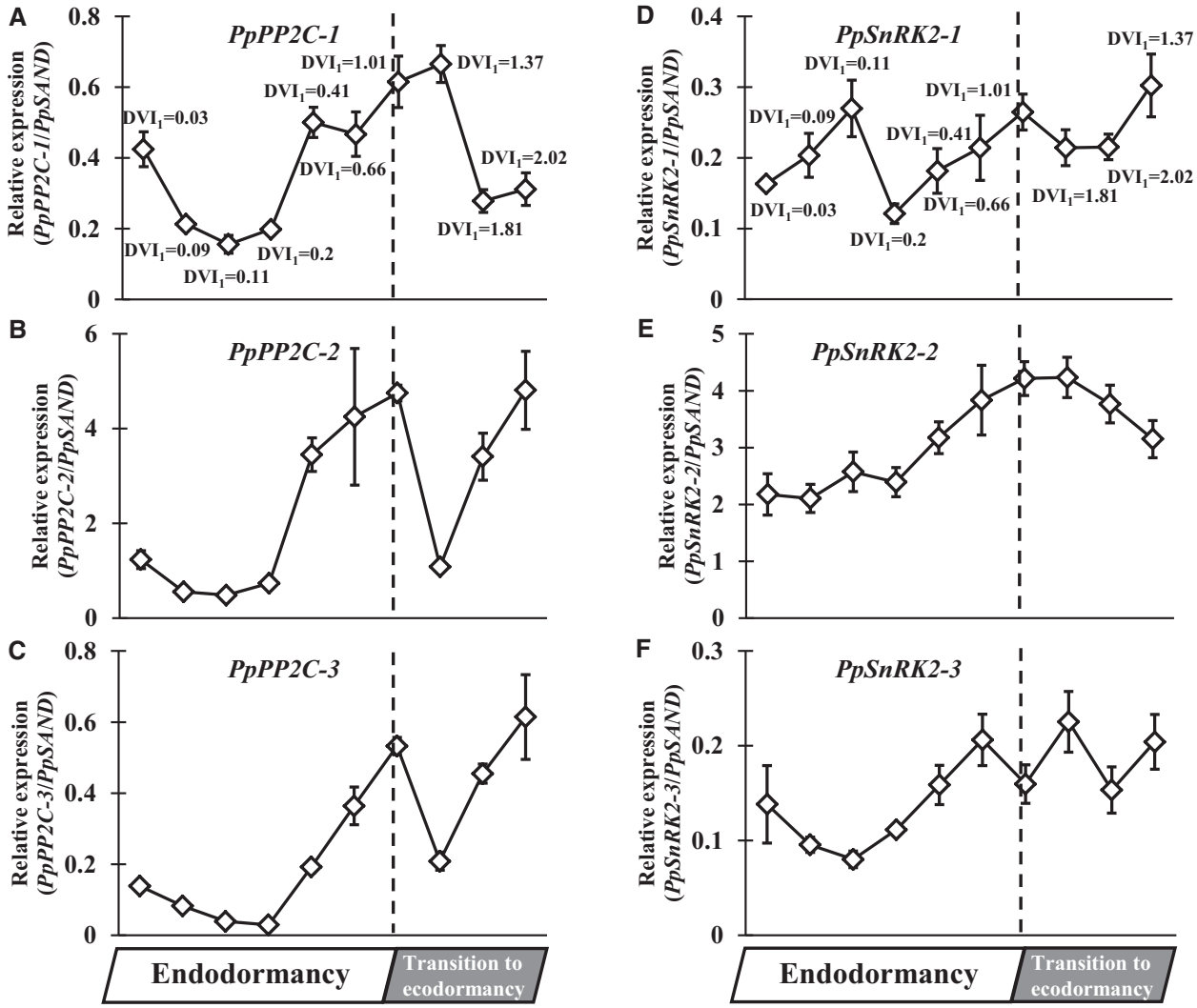


Fig. 3 Expression levels of *PpPP2C* genes (A–C) and *PpSnRK2* genes (D–F) during seasonal endodormancy in 'Kosui' lateral flower buds. Expression values were normalized to the value of the *PpSAND* gene. Each value represents the mean ± SE of three biological replicates, each with three technical replicates. Sampling dates are shown in **Table 1**. Dashed lines indicate the date corresponding to endodormancy release.

As reported in our previous study, the initial bud break in the HC-treated plants was observed 3 weeks after treatment, which was 1 week earlier than the control plants treated with water (Tuan et al. 2016). The earlier bud break of HC-treated plants indicated that HC accelerated the dormancy release of lateral flower buds. The expression level of *PpDAM1* dramatically decreased in both water- and HC-treated plants after transfer to the warm greenhouse (**Fig. 4A**). The transcription level of *PpNCED3* in both the water- and HC-treated plants decreased just after treatment before increasing again after 2 weeks (HC-treated plants) or 3 weeks (water-treated plants). *PpNCED3* expression decreased to a greater extent in HC-treated plants compared with water-treated plants (**Fig. 4B**). *PpABA8′OH*, *PpPP2C-1* and *PpSnRK2-3* shared similar expression patterns during the forcing in that their transcription levels increased and reached the highest levels a week earlier in HC-treated plants compared with water-treated plants (**Fig. 4C, D, I**). mRNA levels of *PpPP2C-2*, *PpPP2C-3*, *PpSnRK2-1* and *PpSnRK2-2* were up-regulated during the forcing; however, no noticeable expression difference between water-treated plants and HC-treated plants was observed (**Fig. 4E–H**). Taken together, the peak expression levels of *PpNCED3*, *PpABA8′OH*, *PpPP2C-1* and *PpSnRK2-3* in HC-treated plants were 1 week earlier than those in water-treated plants, consistent with the 1 week earlier start of bud break in HC-treated plants compared with water-treated plants.

Investigation of the association between PpDAM1 and the *PpNCED3* promoter

As a MADS box transcription factor, DAM is believed to control its target genes by binding to CArG boxes in the promoter regions (Lee et al. 2007, Hao et al. 2015). Based on motif discovery analysis using the PLACE database (http://www.dna.affrc.go.jp/PLACE/), we found a CArG box in the promoter region of *PpNCED3* (nucleotides −624 to −615 relative to the ATG start codon; **Fig. 5A**). To characterize the ability of PpDAM1 to regulate the *PpNCED3* promoter, transient expression of a firefly

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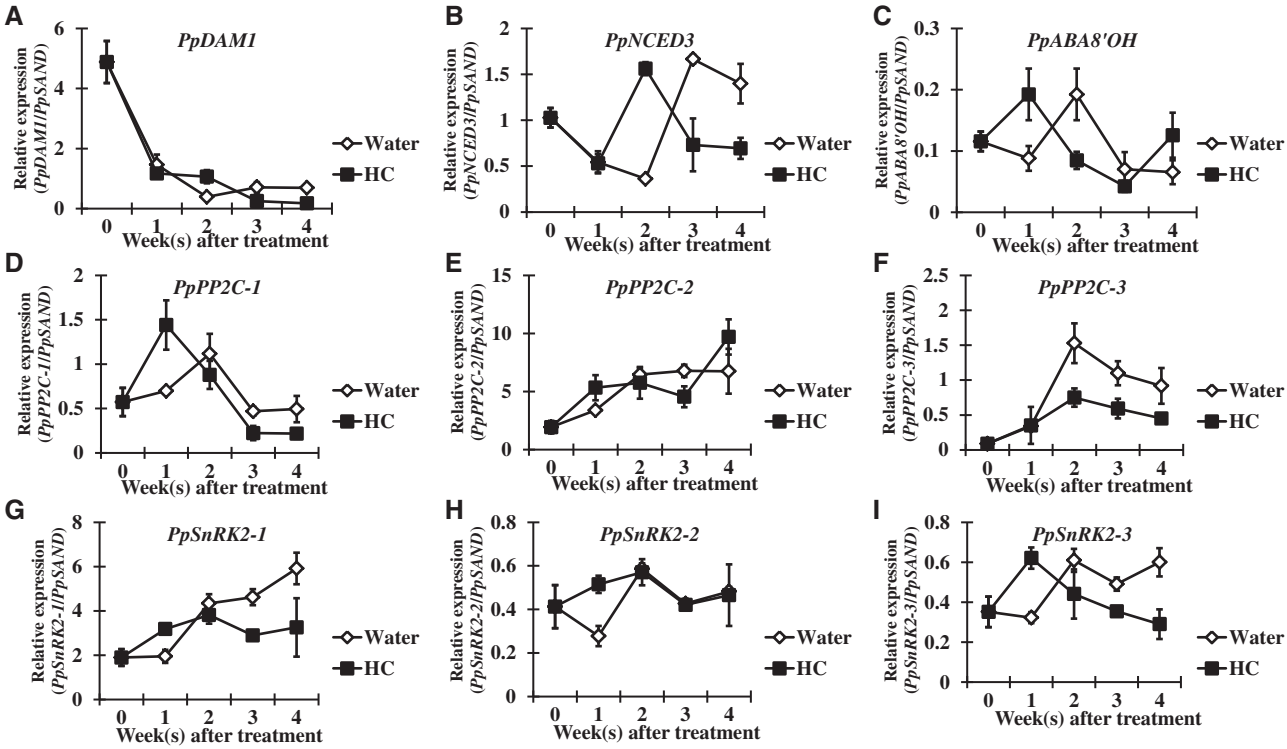


Fig. 4 Effect of HC on expression levels of *PpDAM1* (A), *PpNCED3* (B), *PpABA8′OH* (C), *PpPP2C* genes (D–F), and *PpSnRK2* genes (G–I) in 'Kosui' lateral flower buds. Expression values were normalized to the value of the *PpSAND* gene. Each value represents the mean ± SE of three independent potted plants (biological replicates) with three technical replicates.

luciferase (FLUC) reporter gene under the control of the putative *PpNCED3* promoter regulated by PpDAM1 was evaluated in *Nicotiana benthamiana* leaves. The FLUC activity in the tobacco leaves infiltrated with *Agrobacterium tumefaciens* (*Rhizobium radiobactor*) harboring 35S:PpDAM1 and *PpNCED3* promoter:FLUC vectors was significantly higher than that in the control leaves co-transformed by 35S:empty and *PpNCED3* promoter:FLUC vectors (**Fig. 5B**; Supplementary Fig. S3). This higher FLUC level proved that *PpNCED3* promoter activity was up-regulated by PpDAM1 protein.

To examine further the interaction between PpDAM1 protein and the *PpNCED3* promoter, EMSA was performed. A 40 bp fragment containing a CArG box was used as the probe for EMSA (**Fig. 5A**). A bandshift was observed with labeled probe incubated with PpDAM1 recombinant protein, whereas this bandshift disappeared when a 100-fold competitor was added (**Fig. 5C**). These results indicated that PpDAM1 protein could bind to the *PpNCED3* promoter probe.

Possible association between PpAREB1 and the *PpDAM1* promoter

An increased ABA level possibly due to the activation of *PpNCED3* by PpDAM1 could be a signal for the activation of *AREB/ABF* genes. Interestingly, three ABRE motifs (ACGTG) were found in the putative promoter region of *PpDAM1* (**Fig. 6A**), supporting our hypothesis that the *PpDAM1* promoter is regulated by AREB/ABF transcription factors. Among the several AREB/ABF transcription factors identified in the *P. pyrifolia* transcriptome databases (Liu et al. 2012, Bai et al. 2013), we found one ABRE transcription factor gene (*PpAREB1*) showing high expression during endodormancy. The ORF of *PpAREB1* encodes a protein of 399 amino acids and contains a basic leucine zipper (bZIP) domain in the C-terminus that is conserved in other AREB/ABF transcription factors (Supplementary Fig. S4) (Fujita et al. 2013). The best hit for *PpAREB1* in The Arabidopsis Information Resource (TAIR) GenBank is *AtAREB1* (AT1G45249.3), which was found to be involved in the ABA signaling pathway in response to stresses (Li et al. 2013, Yoshida et al. 2015). The transcription level of *PpAREB1* was then measured in 'Kosui' lateral flower buds during endodormancy and transition stage to ecodormancy (**Fig. 6B**). The expression level of *PpAREB1* was low from $DVI_1 = 0.03$ to $DVI_1 = 0.2$, then increased toward endodormancy release, and decreased to lower levels in the transition stage to ecodormancy.

To test whether PpAREB1 was a functional transcriptional regulator of the *PpDAM1* promoter, a transient LUC assay was performed using *N. benthamiana* leaves. Since the promoter of *PpDAM1* was weakly activated in the leaves of *N. benthamiana*, the density number of FLUC expression calculated by the ChemiDoc™ MP System was affected by the high background noise, leading to large variations. However, the images from the ChemiDoc™ MP System and Student's *t*-test analysis showed that the FLUC activities in the leaves injected with 35S:PpAREB1 and *PpDAM1* promoter:FLUC vectors were significantly lower than those in the leaves injected with 35S:empty and *PpDAM1* promoter:FLUC vectors (**Fig. 6C**; Supplementary Fig. S5A). The activities of the internal control

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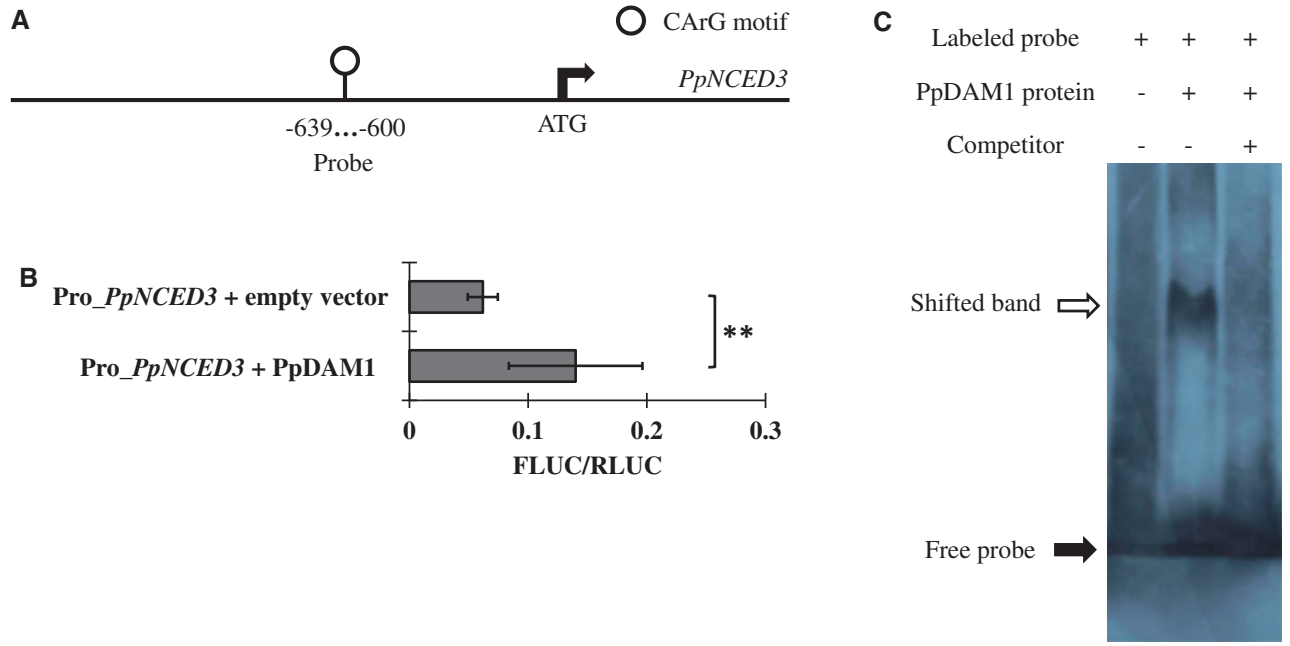


Fig. 5 PpDAM1 up-regulated expression of *PpNCED3*. (A) 5′-Upstream regions of *PpNCED3* genes isolated from 'Kosui' DNA, and location of the CArG motif and the probe used for EMSA in (C). '-n' indicatse the nucleotide position relative to the start codon ATG. (B) Transient activation of the upstream region of *PpNCED3* by PpDAM1 protein. Each value represents the mean ± SE of 10 independent biological replicates. The double asterisk indicates a significant difference of $P < 0.01$ by using the two-tailed Student's *t*-test. (C) EMSA of the PpDAM1 protein binding to the labeled probe (shown in A) in the promoter of *PpNCED3*. The '+' and '—' signs indicate the presence and absence of the corresponding probe or protein, respectively. Open and filled arrows indicate the protein–DNA probe complex and free probes, respectively.

Renilla luciferase (RLUC) did not show large differences between them (Supplementary Fig. S5B). In other words, the activity of the *PpDAM1* promoter was inhibited by the PpAREB1 protein. EMSA was employed to confirm further the association between PpAREB1 protein and the *PpDAM1* promoter in vitro. Three fragments containing ABRE motifs in the *PpDAM1* promoter were chosen for this assay (**Fig. 6A**). The results revealed that the mobility rate of all three fragments containing ABRE motifs was delayed in the presence of PpAREB1 recombinant protein (**Fig. 6D**). The binding signals were inhibited in the presence of a 100-fold competitor. These EMSA results demonstrated that PpAREB1 protein directly binds to the three ABRE motifs in the promoter region of *PpDAM1*.

Discussion

In our previous study, we reported that *PpDAM1* plays a specific role in dormancy phase transition in 'Kosui' lateral leaf buds, where its expression increased toward endodormancy establishment and decreased concomitant with dormancy release (Saito et al. 2013). We were then interested in investigating how the expression of *PpDAM1* changed with the different phases of dormancy. Our previous study that uncovered the factors regulating PpDAM1 in pear showed that the C-repeat binding factor (CBF) is one of the upstream genes that regulates *PpDAM1* expression (Saito et al. 2013). The result was subsequently re-confirmed in Chinese pear (Niu et al. 2016). CBF transcription factors are involved in low temperature signaling pathways in plants and have been considered important regulators in the bud dormancy process (Welling and Palva 2008, Wisniewski et al. 2015, Niu et al. 2016). Overexpression of a peach *CBF* transcription factor in apple altered the expression of *DAM* and other dormancy-related genes, causing delayed bud break in the spring (Wisniewski et al. 2015). Therefore, we assumed that the cooler temperature in late autumn and early winter might trigger endodormancy establishment. With regards to the downstream target genes of DAM, the *FT* gene, which plays an essential role in the transition from vegetative to reproductive growth, is a possible candidate as it has been shown to be a target gene of DAM during bud dormancy (Shim et al. 2014). SVP, a homolog of DAM, has also been shown to repress *FT* in Arabidopsis (Lee et al. 2007). From previous reports, it seems likely that *FT* expression is co-ordinately down-regulated with the induction of *DAM* gene expression, leading to growth cessation during endodormancy establishment in leafy spurge and pear (Horvath et al. 2008, Shim et al. 2014, Hao et al. 2015, Niu et al. 2016), whereas we could not find such an association in Japanese pear (Saito et al. 2015). Therefore, we intended to find other downstream genes of PpDAM1 apart from *FT*. To this end, we screened promoter regions of some candidate genes and found that the putative promoter of *PpNCED3* contains a CArG box, which is identified to be the binding motif of DAM protein (Lee et al. 2007, Hao et al. 2015), suggesting the possibility of the association of PpDAM1 with the *PpNCED3* gene. There are many reports on the possible involvement of ABA in dormancy. In seeds, the

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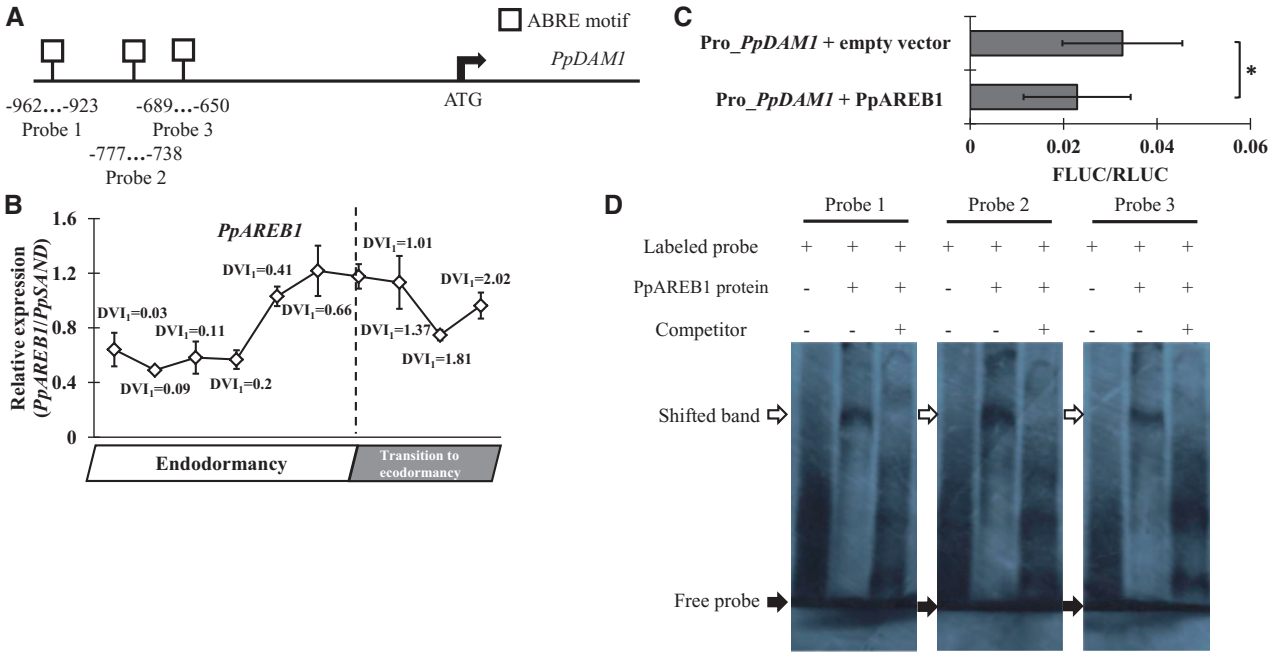


Fig. 6 *PpDAM1* expression was inhibited by PpAREB1. (A) 5′-Upstream regions of *PpDAM1* genes isolated from 'Kosui' DNA, and location of ABRE motifs and probes used for EMSA in (D). '-n' indicates the nucleotide position relative to the start codon ATG. (B) Expression level of *PpAREB1* during seasonal endodormancy in 'Kosui' lateral flower buds. Expression values were normalized to the value of the *PpSAND* gene. Each value represents the mean ± SE of three biological replicates, each with three technical replicates. Sampling dates are shown in **Table 1**. The dashed line indicates the date corresponding to endodormancy release. (C) Transient suppression of the upstream region of *PpDAM1* by PpAREB1 protein. Each value represents the mean ± SE of 22 independent biological replicates. The asterisk indicates a significant difference of $P < 0.05$ by using the two-tailed Student's *t*-test. (D) EMSA of the PpAREB1 protein binding to the labeled probes (shown in A) in the promoter of *PpDAM1*. The '+' and '—' signs indicate the presence and absence of the corresponding probe or protein, respectively. Open and filled arrows indicate the protein–DNA probe complex and free probes, respectively.

important role of ABA in the induction and maintenance of dormancy has been shown with respect to interaction with other hormones and their associated signaling networks (Kermode 2005, Graeber et al. 2012). For example, it is clear that a dynamic balance between ABA and gibberellins controls the seed dormancy and germination (Kermode 2005, Graeber et al. 2012). Although ABA is well known as an important hormone for bud dormancy, the details of the regulation of ABA biosynthesis during bud dormancy remain unclear. In the present study, we found that ABA content increased with the progression of endodormancy and dropped during the transition stage from endodormancy to ecodormancy in 'Kosui' lateral flower buds (**Fig. 2A**). Expression levels of ABA metabolism- and its signaling-related genes were also up-regulated concomitant with endodormancy release (**Figs. 2B, C**, **3**). Application of ABA inhibits dormancy release in grape buds (Zheng et al. 2015, Li and Dami 2016), and delays bud break in apple (Guak and Fuchigami 2001) and birch (Li et al. 2003). Therefore, we assume that these increases in ABA content and expression of ABA metabolism- and its signaling-related gene are involved in the endodormancy status. ABA metabolism- and its signaling-related genes are probably inhibited when lateral flower buds meet the required chilling accumulation to complete endodormancy, followed by their transition stage to the ecodormancy. The high expression of the ABA catabolism gene, *PpABA8′OH*, during the transition phase to ecodormancy may be responsible for the decrease of the ABA level (**Fig. 2A, C**). In grape, ABA accumulation and *NCED* expression rise during dormancy induction, reach their highest levels at the dormancy maintenance stage and gradually decrease with dormancy release (Zheng et al. 2015). Similar trends in ABA level and *NCED* expression have also been reported in peach (Wang et al. 2016) and poplar (Rohde et al. 2002, Arora et al. 2003, Rohde et al. 2007, Ruttink et al. 2007), with the peak expression levels detected during the mid dormancy process.

PpNCED3, *PpPP2C-2*, *PpPP2C-3*, *PpSnRK2-1* and *PpSnRK2-3* expression decreased after endodormancy release but they were then induced in the late part of transition stage to the ecodormancy (**Figs. 2B**; **3B–D, F**). Moreover, the peak expression level of *PpNCED3*, *PpABA8′OH*, *PpPP2C-1* and *PpSnRK2-3* in HC-treated plants was 1 week earlier than that in water-treated plants (**Fig. 4B, C, D, I**). This suggested that the increases in their expression levels were associated with the time point of dormancy release. Expression levels of *PpNCED3* were increased in the late part of transition stage to the ecodormancy or after 2 weeks in HC-treated plants and 3 weeks in water-treated plants (**Figs. 2B**, **4B**), when very low levels of *PpDAM1* were observed (**Figs. 1A**, **4A**). This suggested that PpDAM1 did not regulate *PpNCED3* expression during these stages.

The results of the transient LUC assay and EMSA showed that PpDAM1 activated *PpNCED3* expression by binding to the

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CArG motif in the *PpNCED3* promoter (**Fig. 5B, C**). However, the time of increase of *PpNCED3* expression lagged behind that of *PpDAM1* during seasonal endodormancy; *PpNCED3* expression only started to increase at $DVI_1 = 0.41$, when *PpDAM1* mRNA accumulation was at its highest level (**Figs. 1A**, **2B**). Based on these findings, we hypothesize that *PpNCED3* is not regulated exclusively by *PpDAM1*, but other factors are also involved in its regulation. Among all the five *NCED* genes of Arabidopsis, *PpNCED3* shares the highest identity with *AtNCED3* (Supplementary Fig. S2B), which has been shown to increase in response to dehydration (Iuchi et al. 2001, Tan et al. 2003). Therefore, *PpNCED3* may also be regulated by drought stress, which occurs in the winter season. In addition, as a MADS box transcription factor, PpDAM1 protein probably forms a protein complex with other transcription factors and chromatin-associated proteins to regulate *PpNCED3* (Smaczniak et al. 2012). *Prunus mume DAM* genes (*PmDAM*) were shown to interact with the homolog of Arabidopsis SOC1 (PmSOC1) by a yeast two-hybrid screening, in which synchronized expression patterns of *PmDAM6* and *PmSOC1* were reported during dormancy release in flower buds (Kitamura et al. 2016). Therefore, we inferred, in the absence of the co-operating proteins, that PpDAM1 protein alone probably cannot activate the activity of the *PpNCED3* promoter during establishment of endodormancy. Our results from the transient LUC assay showed that PpDAM1 protein may successfully find endogenous proteins, possibly tobacco-specific SOC1 proteins, to form a functional complex and may up-regulate the activity of *PpNCED3* promoters (**Fig. 5B**). In seeds, the *DELAY OF GERMINATION 1* (*DOG1*) gene and ABA are major factors for the induction of seed dormancy (Graeber et al. 2012). Recently, increasing evidence has shown a common regulatory network in seed and bud dormancy (Leida et al. 2012, Wu et al. 2015, Wang et al. 2016), but we could not find a gene showing high similarity to *DOG1* in the pear genome. Our results showed that endodormancy establishment was correlated with the induction of the *PpDAM1* gene, which may also be directly involved in ABA metabolism and signaling pathways. In addition, Arabidopsis MYB96 has also been identified as controlling the activity of *NCED* genes, contributing to the fine-tuning of seed dormancy regulation (Lee et al. 2015). In our pear database, PpMYB306 (Pbr011441.1) showed the highest similarity to MYB96, albeit that only a $< 50\%$ homology with MYB96 was observed (Bai et al. 2013). Interestingly, the promoter region of *PpMYB306* contains an ABRE-binding motif and we found an increased expression of *PpMYB306* towards endodormancy release (unpublished data), leading to the inference that this gene might have some role in endodormancy via the ABA signaling pathway. Briefly, this is the first evidence that PpDAM1, a positive regulator of bud dormancy, interacts with an upstream element of the *NCED* promoter and positively controls its expression. This is in contrast to our expectation that PpDAM1 is a repressive transcription factor due to the ethylene-responsive element-binding factor-associated amphiphilic repression (EAR) motif at its C-terminus (Supplementary Fig. S1) (Kagale and Rozwadowski 2011). Hill et al. (2008) showed that a MADS transcription factor, AGL15, containing an EAR motif induces expression of some direct downstream target genes, and represses the expression of others, suggesting that protein interactions determine whether a protein acts as a transcriptional activator or repressor. It is therefore likely that DAMs can also act as either positive or negative regulators, which depends on their co-operating proteins.

As discussed, *DAM* and ABA are both well known as important factors regulating bud dormancy; therefore, it is interesting to determine the connection between DAM and ABA signaling-related genes. Here, we reported an ABRE-binding transcription factor, PpAREB1, which can bind to three ABRE motifs in the promoter region of *PpDAM1*, negatively regulating its activity (**Fig. 6C, D**). This is in accordance with recent reports in which AREBs are found to function as negative regulators. For example, an AREB1 transcription factor forms a complex with a NAC2 transcription factor and negatively regulates the expression of *NCED1* in peanut (Liu et al. 2016). It is proposed that *PpDAM1* expression is activated by the CBF transcription factor in endodormancy establishment (Saito et al. 2015), while our results show that the increased level of *PpAREB1* expression may be responsible for the reduced expression of *PpDAM1* in endodormancy release of pear buds directly or indirectly by the association with other transcription factor such as NAC (**Figs. 1A**, **6B**). Besides these, the involvement of microRNAs or histone modification to *PpDAM1* was also reported in dormancy regulation (Saito et al. 2015, Niu et al. 2016). In the absence of a mechanism for down-regulation of *DAM*, the buds remain in the dormant state without sprouting, as shown in the *PmDAM6*-overexpressing poplar plant (Sasaki et al. 2011). On the other hand, by RNA sequencing analysis during dormancy (Bai et al. 2013), we have identified many genes possessing the ABRE-binding motif, showing significant changes in gene expression in endodormancy. For example, expression of the ABC transporter gene, which is essential for plant growth and development, was induced toward endodormancy release (Kang et al. 2011), while that of α- and β-expansin genes, which are related to plant cell growth, was repressed (Marowa et al. 2016). Although there was no information as to whether these genes are under the regulation of AREB1, AREB1 could possibly regulate many downstream genes to optimize endodormany release. Further studies are needed to clarify this possibility. Indeed, we also found ABRE motifs in the promoter regions of other genes such as those encoding CBFs and FTs that are believed to have functional roles in bud dormancy in pear. This suggests the possibility of a large signaling network of AREB/ABF transcription factors operating during bud dormancy. Compared with seed dormancy, studies on the regulation by AREB/ABF transcription factors of bud dormancy are still limited. The possibility of interaction between *DAM* genes and the ABA-insensitive (ABI) transcription factor during bud dormancy of pear was discussed by Niu et al. (2016). Studies showed that the ectopic expression of the mutant Arabidopsis *abi1* gene modified the dormancy response of lateral buds to exogenous ABA, where exogenous ABA maintained lateral bud dormancy in wild-type poplar but not in *abi1* transgenic poplar (Arend et al. 2009). Similarly, down-regulation of *ABI5* resulted in an early sprouting phenotype in dormant cormels, revealing the role of *ABI5* in bud dormancy (vegetative organ) in

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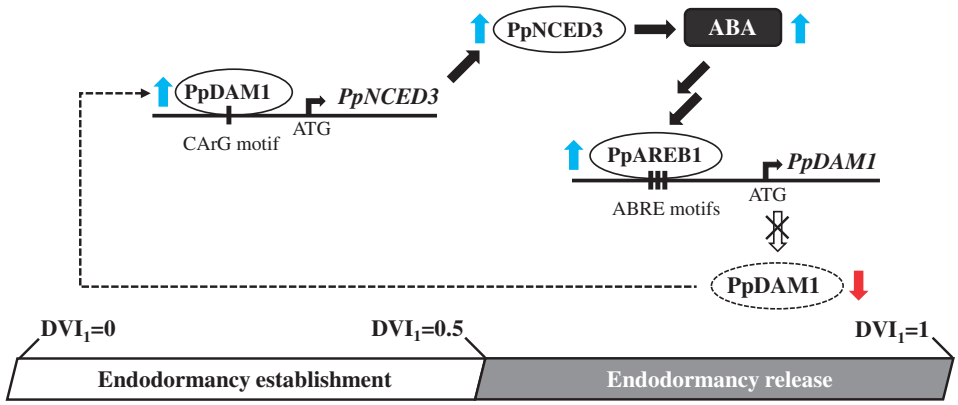


Fig. 7 Schematic diagram showing the proposed interaction between *PpDAM1* and the ABA metabolism and signaling pathway during endodormancy in 'Kosui' lateral flower buds.

Gladiolus (Wu et al. 2015). Based on our results, we propose that *PpAREB1* is a candidate gene regulating bud dormancy in pear. We also investigated the expression of *PpDAM1*, *PpNCED3* and *PpAREB1* in the lateral flower buds of low-chill Taiwanese pear 'Hengshanli', a less dormant cultivar without distinct endodormancy (Supplementary Fig. S6). The results showed that *PpDAM1* expression increased earlier than that of *PpNCED3* and *PpAREB1* in the 'Hengshanli' lateral flower buds, although the high expression of *PpNCED3* in the early bud dormancy stage needs to be clarified. Taken together, all these results suggested a common co-operative mechanism of *PpDAM1*, *PpNCED3* and *PpAREB1* in lateral flower buds of pear.

Conclusion

We here report the potential role of *PpDAM1* in endodormancy establishment of 'Kosui' lateral flower buds. We also show that there is ABA content and increased expression of genes involved in ABA metabolism and signaling pathways with endodormancy release. Specifically, PpDAM1 activates the expression of *PpNCED3*, while *PpDAM1* transcription is inhibited by PpAREB1, suggesting a feedback regulation during endodormancy (**Fig. 7**). In this model, *PpDAM1* expression is first induced during endodormancy establishment. The higher level of *PpDAM1* activates the promoter activities of *PpNCED3*, leading to the high expression of *PpNCED3* during endodormancy release. Following this, ABA is synthesized, initiating ABA catabolism- (*PpABA8′OH*) and signaling- (*PpPP2C* genes and *PpSnRK2* genes) related gene expression. The higher transcription of *PpSnRK2* genes results in up-regulated expression of *PpAREB1*, which later causes the decreased expression of *PpDAM1* during endodormancy release. We also have to consider the regulation of *PpDAM1* by epigenetic factors such as histone modification and methylation (Saito et al. 2013, Saito et al. 2015), which have not been indicated in the current model proposed in **Fig. 7**. This is the first evidence of a direct interaction between a *DAM* gene and ABA metabolism and signaling pathways in vitro. These findings will provide valuable information to uncover the regulatory mechanism of *DAM* genes in vivo during bud endodormancy of pear in particular and of deciduous trees in general.

Materials and Methods

Plant materials

All lateral flower buds were collected from Japanese pears (*Pyrus pyrifolia* Nakai. 'Kosui') grown at the Institute of Fruit Tree and Tea Science, NARO, Tsukuba, Japan (36°N, 140°E). Samples were collected between 08:00 and 09:00 h.

Hydrogen cyanamide (HC) treatment

In mid winter (December 22, 2014), 1% HC solution (w/v) or water was applied to lateral flower buds of 3-year-old nursery trees of 'Kosui' grafted on *P. calleryana* Decne (purchased from Komachi-En, Co., Ltd.) grown in natural conditions in the orchard of the Institute of Fruit Tree and Tea Science, NARO. After treatment, the potted trees were transferred to a greenhouse and maintained at a minimum temperature of 10°C. For gene expression analysis, lateral flower buds were harvested from HC- and water-treated trees at 1, 2, 3 and 4 weeks after treatment, and then immediately frozen in liquid nitrogen, and stored at −80°C until RNA extraction. Three single 'Kosui' trees in three independent pots were used for each treatment and at least 10 flower buds were collected from each tree at approximately the same time of the day for one biological replicate.

Quantitative real-time reverse transcription–PCR (qRT–PCR)

For expression analysis, we used the lateral flower buds of newly grown long shoots without secondary elongation in the season of 2015–2016. Lateral flower buds were collected in three biological replicate pools, and immediately frozen in liquid nitrogen and then stored at −80°C until analysis. The sampling dates and DVI value in each sampling date are shown in **Table 1**. Total RNA was isolated from lateral flower buds of 'Kosui' as previously described by Wan and Wilkins (1994). First-strand cDNA synthesis was performed using the SuperScript™ III First strand synthesis system (Invitrogen). A 1 µg aliquot of RNA was first treated with DNase (Promega) and reverse-transcribed using SuperScript III oligo(dT)$_{20}$ primers according to the manufacturer's instructions (Invitrogen). Specific primers were designed for quantitative real-time PCR (Supplementary Table S2). The real-time quantification of the first strand cDNA was performed using the 7500 Real Time PCR System (Applied Biosystems) and analyzed with 7500 Software ver. 2.0 (Applied Biosystems). For the control reaction, no template was added to the reaction mixture, resulting in no detectable fluorescent signal. The PCR conditions were set as follows: initial denaturation for 10 s at 95°C, followed by 40 cycles of denaturation at 95°C for 5 s, and annealing at 60°C and extension for 34 s at 72°C. Three biological replicates with three technical replicates were used for each sample.

Western blot analysis

Lateral flower buds collected from 'Kosui' plants on November 7, 2015 (DVI$_1$ = 0.09), December 8, 2015 (DVI$_1$ = 0.41) and January 19, 2016

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(DVI_1 = 1.37) were used to check expression of the PpDAM1 protein. Total proteins were extracted using the SDS extraction method described by Song et al. (2006). Total extracted proteins were then stained with Coomassie brilliant blue for documentation of the amount of protein used in the Western blot (Supplementary Fig. S7). After separation on a 14% SDS–polyacrylamide gel (e-PAGEL; ATTO) and transfer to Hybond-P membranes (Amersham Biosciences), PpDAM1 protein was detected using a WesternBreeze™ kit (Invitrogen) with an antibody-recognizing peptide sequence of PpDAM1 (CSDTLSLKLGLP), provided by Scrum.

Transient LUC assay

The putative promoter region of *PpNCED3*, which is approximately 2,000 bp upstream from the start codon, was isolated from 'Kosui' using the draft genome information of 'Suli' pear (*P. bretschneideri* Rehd.) (Wu et al. 2013) (primers are listed in Supplementary Table S2). The promoter region and ORF of *PpDAM1* were isolated from 'Kosui' as shown in our previous reports (Saito et al. 2013, Saito et al. 2015). The putative promoter regions of *PpNCED3* and *PpDAM1* were then cloned into pGWB35 (Nakagawa et al. 2007), in which the FLUC gene is under the control of the cloned promoters (primers are listed in Supplementary Table S2). The ORFs of *PpDAM1* and *PpAREB1* were amplified from first-strand cDNA of 'Kosui' and cloned into the pGWB2 vector under transcriptional control of the 35S *Cauliflower mosaic virus* (CaMV) promoter. As the internal standard vector, the RLUC gene amplified from the pRL-SV40 vector (Promega) was cloned into the pBGW2 vector. All recombinant plasmids were then transferred into *A. tumefaciens* EHA105 and transient expression assays were carried out using *N. benthamiana* leaves with the method available in our laboratory (Bai et al. 2014, Tuan et al. 2015). Five days after infiltration, the infiltrated leaf discs were harvested for FLUC and RLUC assay using dual luciferase assay reagents (Promega), and then FLUC and RLUC activities were detected using the ChemiDoc™ MP System (Bio Rad). The data are represented as the ratio of FLUC to RLUC activity (FLUC/RLUC). A total of 10 and 22 independent biological replicates (different infiltrated leaf discs) were used to investigate PpDAM1 protein–*PpNCED3* promoter and PpAREB1 protein–*PpDAM1* promoter interactions, respectively.

Electrophoretic mobility shift assay (EMSA)

For protein synthesis, ORFs of *PpDAM1* and *PpAREB1* were amplified from first-strand cDNA of 'Kosui' and cloned into pET-32a (+) vector (Novagen). The resultant constructs were then transferred to *Escherichia coli* BL21 (DE3). Protein expression was induced by isopropyl-β-D-thiogalactoside (IPTG) at a final concentration of 1 mM as described by Liu and Yang (2012), and the proteins were purified using a HisTALON™ Gravity Column Purification Kit (Clontech). For EMSA, a 40 bp fragment of the *PpNCED3* promoter containing the CArG motif and three 40 bp fragments of the *PpDAM1* promoter containing ABRE motifs were used as probes. EMSA assay was carried out using the DIG Gel Shift Kit, 2nd Generation (Roche Diagnostics), according to the manufacturer's instructions.

Measurement of ABA levels

Lateral flower buds were collected from 'Kosui' in the season of 2014–2015 (the sampling dates and DVI value in each sampling dates are shown in **Table 1**). Approximately 0.5 g samples per extraction were used to analyze ABA content using ultra-performance liquid chromatography/mass spectrometry/mass spectrometry (UPLC/MS/MS). Extraction and purification of ABA was performed as described by Dobrev and Kamínek (2002) with slight modifications. Frozen samples (approximately 0.5 g) were homogenized in liquid nitrogen with a mortar and pestle, mixed with a solution of methanol/water/formic acid (15:4:1) containing internal standard (10 ng of [$^2H_6$]ABA, OlchemIm Ltd.), and then extracted at −20°C overnight. The extract was purified following the method of Dobrev and Kamínek (2002). The acidic fraction that contained ABA was evaporated to dryness. The dried samples were then dissolved in 3% aqueous acetonitrile, and injected into a UPLC/MS/MS system consisting an UPLC (AQUITY UPLC; Waters Corporation) and a tandem quadrupole mass detector (AQUITY TQ Detector, Waters Corporation). ABA was separated using an ODS column (AQUITY UPLC BEH C_{18}, 1.7 µm, 2.1 mm × 100 mm, Waters Corporation) at a flow rate of 0.2 ml min^{-1} with gradients of acetonitrile/water containing 0.05% acetic acid. ABA was quantified in selected ion recording mode with the internal standard method using the deuterated standard. Three biological replicates with three technical replicates were used for each sample.

Supplementary data

Supplementary data are available at PCP online.

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Disclosures

The authors have no conflicts of interest to declare.

References

Andersen, C.L., Jensen, J.L. and Ørntoft, T.F. (2004) Normalization of real-time quantitative reverse transcription–PCR data: a model-based variance estimation approach to identify genes suited for normalization, applied to bladder and colon cancer data sets. *Cancer Res.* 64: 5245–5250.

Arend, M., Schnitzler, J.-P., Ehlting, B., Hänsch, R., Lange, T., Rennenberg, H., et al. (2009) Expression of the Arabidopsis mutant *abi1* gene alters abscisic acid sensitivity, stomatal development, and growth morphology in gray poplars. *Plant Physiol.* 151: 2110–2119.

Arora, R., Rowland, L.J. and Tanino, K. (2003) Induction and release of bud dormancy in woody perennials: a science comes of age. *Hortscience* 38: 911–921.

Atkinson, C.J., Brennan, R.M. and Jones, H.G. (2013) Declining chilling and its impact on temperate perennial crops. *Environ. Exp. Bot.* 91: 48–62.

Bai, S., Saito, T., Honda, C., Hatsuyama, Y., Ito, A. and Moriguchi, T. (2014) An apple B-box protein, MdCOL11, is involved in UV-B- and temperature-induced anthocyanin biosynthesis. *Planta* 240: 1051–1062.

Bai, S., Saito, T., Sakamoto, D., Ito, A., Fujii, H. and Moriguchi, T. (2013) Transcriptome analysis of Japanese pear (*Pyrus pyrifolia* Nakai) flower buds transitioning through endodormancy. *Plant Cell Physiol.* 54: 1132–1151.

Bielenberg, D., Wang, Y., Li, Z., Zhebentyayeva, T., Fan, S., Reighard, G., et al. (2008) Sequencing and annotation of the evergrowing locus in peach [*Prunus persica* (L.) Batsch] reveals a cluster of six MADS-box transcription factors as candidate genes for regulation of terminal bud formation. *Tree Genet. Genomes* 4: 495–507.

Böhlenius, H., Huang, T., Charbonnel-Campaa, L., Brunner, A.M., Jansson, S., Strauss, S.H., et al. (2006) *CO/FT* regulatory module controls timing of flowering and seasonal growth cessation in trees. *Science* 312: 1040–1043.

Cooke, J.E.K., Eriksson, M.E. and Junttila, O. (2012) The dynamic nature of bud dormancy in trees: environmental control and molecular mechanisms. *Plant Cell Environ.* 35: 1707–1728.

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Crabbe, J. and Barnola, P. (1996) A new conceptual approach to bud dormancy in woody plants. *In* Plant Dormancy: Physiology, Biochemistry and Molecular Biology. Edited by Lang, G.A. pp. 83–113. CAB International, London.

Dobrev, P.I. and Kamínek, M. (2002) Fast and efficient separation of cytokinins from auxin and abscisic acid and their purification using mixed-mode solid-phase extraction. *J. Chromatogr. A* 950: 21–29.

Finkelstein, R. (2013) Abscisic acid synthesis and response. *Arabidopsis Book* 11: e0166.

Fujita, Y., Yoshida, T. and Yamaguchi-Shinozaki, K. (2013) Pivotal role of the AREB/ABF–SnRK2 pathway in ABRE-mediated transcription in response to osmotic stress in plants. *Physiol. Plant.* 147: 15–27.

Graeber, K.A.I., Nakabayashi, K., Miatton, E., Leubner-Metzger, G. and Soppe, W.J.J. (2012) Molecular mechanisms of seed dormancy. *Plant Cell Environ.* 35: 1769–1786.

Guak, S. and Fuchigami, L.H. (2001) Effects of applied ABA on growth cessation, bud dormancy, cold acclimation, leaf senescence and N mobilization in apple nursery plants. *J. Hortic. Sci. Biotechnol.* 76: 459–464.

Hao, X., Chao, W., Yang, Y. and Horvath, D. (2015) Coordinated expression of *FLOWERING LOCUS T* and *DORMANCY ASSOCIATED MADS-BOX*-like genes in leafy spurge. *PLoS One* 10: e0126030.

Hill, K., Wang, H., Perry, S.E. (2008) A transcriptional repression motif in the MADS factor AGL15 is involved in recruitment of histone deacetylase complex components. *Plant J.* 53: 172–185.

Horvath, D.P., Chao, W.S., Suttle, J.C., Thimmapuram, J. and Anderson, J.V. (2008) Transcriptome analysis identifies novel responses and potential regulatory genes involved in seasonal dormancy transitions of leafy spurge (*Euphorbia esula* L.). *BMC Genomics* 9: 1–17.

Howe, G.T., Horvath, D.P., Dharmawardhana, P., Priest, H.D., Mockler, T.C. and Strauss, S.H. (2015) Extensive transcriptome changes during natural onset and release of vegetative bud dormancy in *Populus*. *Front. Plant. Sci.* 6: 989.

Hsu, C.-Y., Adams, J.P., Kim, H., No, K., Ma, C., Strauss, S.H., et al. (2011) *FLOWERING LOCUS T* duplication coordinates reproductive and vegetative growth in perennial poplar. *Proc. Natl. Acad. Sci. USA* 108: 10756–10761.

Ibáñez, C., Kozarewa, I., Johansson, M., Ögren, E., Rohde, A. and Eriksson, M.E. (2010) Circadian clock components regulate entry and affect exit of seasonal dormancy as well as winter hardiness in *Populus* trees. *Plant Physiol.* 153: 1823–1833.

Imai, T., Ubi, B.E., Saito, T. and Moriguchi, T. (2014) Evaluation of reference genes for accurate normalization of gene expression for real time-quantitative PCR in *Pyrus pyrifolia* using different tissue samples and seasonal conditions. *PLoS One* 9: e86492.

Iuchi, S., Kobayashi, M., Taji, T., Naramoto, M., Seki, M., Kato, T., et al. (2001) Regulation of drought tolerance by gene manipulation of 9-*cis*-epoxycarotenoid dioxygenase, a key enzyme in abscisic acid biosynthesis in Arabidopsis. *Plant J.* 27: 325–333.

Jiménez, S., Lawton-Rauh, A.L., Reighard, G.L., Abbott, A.G. and Bielenberg, D.G. (2009) Phylogenetic analysis and molecular evolution of the dormancy associated MADS-box genes from peach. *BMC Plant Biol.* 9: 1–12.

Jiménez, S., Reighard, G.L. and Bielenberg, D.G. (2010) Gene expression of *DAM5* and *DAM6* is suppressed by chilling temperatures and inversely correlated with bud break rate. *Plant Mol. Biol.* 73: 157–167.

Kagale, S. and Rozwadowski, K. (2011) EAR motif-mediated transcriptional repression in plants: an underlying mechanism for epigenetic regulation of gene expression. *Epigenetics* 6: 141–146.

Kang, J., Park, J., Choi, H., Burla, B., Kretzschmar, T., Lee, Y., et al. (2011) Plant ABC transporters. *Arabidopsis Book* 9: e0153.

Kermode, R.A. (2005) Role of abscisic acid in seed dormancy. *J. Plant Growth Regul.* 24: 319–344.

Kitamura, Y., Takeuchi, T., Yamane, H. and Tao, R. (2016) Simultaneous down-regulation of *DORMANCY-ASSOCIATED MADS-box6* and *SOC1* during dormancy release in Japanese apricot (*Prunus mume*) flower buds. *J. Hortic. Sci. Biotechnol.* 91: 476–482.

Kuroki, K., Takemura, Y., Matsumoto, K., Takeda, M., Tomiyama, M. and Tamura, F. (2013) Effect of hydrogen cyanamide on breaking flower bud endodormancy and flowering period of major Japanese pear cultivars. *Hortic. Res. (Japan)* 12: 179–185.

Lang, G.A. (1987) Dormancy: a new universal terminology. *Hortscience* 22: 817–820.

Lee, H.G., Lee, K. and Seo, P.J. (2015) The *Arabidopsis* MYB96 transcription factor plays a role in seed dormancy. *Plant Mol. Biol.* 87: 371–381.

Lee, J.H., Yoo, S.J., Park, S.H., Hwang, I., Lee, J.S. and Ahn, J.H. (2007) Role of *SVP* in the control of flowering time by ambient temperature in *Arabidopsis*. *Genes Dev.* 21: 397–402.

Leida, C., Conejero, A., Arbona, V., Gómez-Cadenas, A., Llácer, G., Badenes, M.L., et al. (2012) Chilling-dependent release of seed and bud dormancy in peach associates to common changes in gene expression. *PLoS One* 7: e35777.

Li, C., Junttila, O., Heino, P. and Palva, E.T. (2003) Different responses of northern and southern ecotypes of *Betula pendula* to exogenous ABA application. *Tree Physiol.* 23: 481–487.

Li, S. and Dami, I.E. (2016) Responses of *Vitis vinifera* 'Pinot gris' grapevines to exogenous abscisic acid (ABA): I. Yield, fruit quality, dormancy, and freezing tolerance. *J. Plant Growth Regul.* 35: 245–255.

Li, X.-Y., Liu, X., Yao, Y., Li, Y.-H., Liu, S., He, C.-Y., et al. (2013) Overexpression of *Arachis hypogaea AREB1* gene enhances drought tolerance by modulating ROS scavenging and maintaining endogenous ABA content. *Int. J. Mol. Sci.* 14: 12827–12842.

Liu, G., Li, W., Zheng, P., Xu, T., Chen, L., Liu, D., et al. (2012) Transcriptomic analysis of 'Suli' pear (*Pyrus pyrifolia* white pear group) buds during the dormancy by RNA-Seq. *BMC Genomics* 13: 1–18.

Liu, S., Li, M., Su, L., Ge, K., Li, L., Li, X., et al. (2016) Negative feedback regulation of ABA biosynthesis in peanut (*Arachis hypogaea*): a transcription factor complex inhibits *AhNCED1* expression during water stress. *Sci. Rep.* 6: 37943.

Liu, Z.Q. and Yang, P.C. (2012) Construction of pET-32 α (+) vector for protein expression and purification. *N. Amer. J. Med. Sci.* 4: 651–655.

Marowa, P., Ding, A. and Kong, Y. (2016) Expansins: roles in plant growth and potential applications in crop improvement. *Plant Cell Rep.* 35: 949–965.

Mohamed, R., Wang, C.-T., Ma, C., Shevchenko, O., Dye, S.J., Puzey, J.R., et al. (2010) *Populus CEN/TFL1* regulates first onset of flowering, axillary meristem identity and dormancy release in *Populus*. *Plant J.* 62: 674–688.

Nakagawa, T., Kurose, T., Hino, T., Tanaka, K., Kawamukai, M., Niwa, Y., et al. (2007) Development of series of gateway binary vectors, pGWBs, for realizing efficient construction of fusion genes for plant transformation. *J. Biosci. Bioeng.* 104: 34–41.

Ng, L.M., Melcher, K., Teh, B.T. and Xu, H.E. (2014) Abscisic acid perception and signaling: structural mechanisms and applications. *Acta Pharmacol. Sin.* 35: 567–584.

Niu, Q., Li, J., Cai, D., Qian, M., Jia, H., Bai, S., et al. (2016) Dormancy-associated MADS-box genes and microRNAs jointly control dormancy transition in pear (*Pyrus pyrifolia* white pear group) flower bud. *J. Exp. Bot.* 67: 239–257.

Or, E., Belausov, E., Popilevsky, I. and Tal, Y.B. (2000) Changes in endogenous ABA level in relation to the dormancy cycle in grapevines grown in a hot climate. *J. Hortic. Sci. Biotechnol.* 75: 190–194.

Oya, Y. (2006) Prediction method for anthesis of Japanese pears based on weather habit reactions. *Bull. Tochigi Agric. Exp. Stn. (Japan)* 58: 7–16.

Rohde, A., Prinsen, E., De Rycke, R., Engler, G., Van Montagu, M. and Boerjan, W. (2002) PtABI3 impinges on the growth and differentiation of embryonic leaves during bud set in poplar. *Plant Cell* 14: 1885–1901.

Rohde, A., Ruttink, T., Hostyn, V., Sterck, L., Van Driessche, K. and Boerjan, W. (2007) Gene expression during the induction, maintenance, and release of dormancy in apical buds of poplar. *J. Exp. Bot.* 58: 4047–4060.

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Ruttink, T., Arend, M., Morreel, K., Storme, V., Rombauts, S., Fromm, J., et al. (2007) A molecular timetable for apical bud formation and dormancy induction in poplar. *Plant Cell* 19: 2370–2390.

Saito, T., Bai, S., Imai, T., Ito, A., Nakajima, I. and Moriguchi, T. (2015) Histone modification and signalling cascade of the *dormancy-associated MADS-box* gene, *PpMADS13-1*, in Japanese pear (*Pyrus pyrifolia*) during endodormancy. *Plant Cell Environ.* 38: 1157–1166.

Saito, T., Bai, S., Ito, A., Sakamoto, D., Saito, T., Ubi, B.E., et al. (2013) Expression and genomic structure of the *dormancy-associated MADS box* genes *MADS13* in Japanese pears (*Pyrus pyrifolia* Nakai) that differ in their chilling requirement for endodormancy release. *Tree Physiol.* 33: 654–667.

Sasaki, R., Yamane, H., Ooka, T., Jotatsu, H., Kitamura, Y., Akagi, T., et al. (2011) Functional and expressional analyses of *PmDAM* genes associated with endodormancy in Japanese apricot. *Plant Physiol.* 157: 485–497.

Schwartz, S.H., Qin, X. and Zeevaart, J.A.D. (2003) Elucidation of the indirect pathway of abscisic acid biosynthesis by mutants, genes, and enzymes. *Plant Physiol.* 131: 1591–1601.

Seo, M. and Koshiba, T. (2002) Complex regulation of ABA biosynthesis in plants. *Trends Plant Sci.* 7: 41–48.

Shim, D., Ko, J.-H., Kim, W.-C., Wang, Q., Keathley, D.E. and Han, K.-H. (2014) A molecular framework for seasonal growth–dormancy regulation in perennial plants. *Hortic. Res.* 1: 14059.

Smaczniak, C., Immink, R.G.H., Muiño, J.M., Blanvillain, R., Busscher, M., Busscher-Lange, J., et al. (2012) Characterization of MADS-domain transcription factor complexes in *Arabidopsis* flower development. *Proc. Natl. Acad. Sci. USA* 109: 1560–1565.

Song, J., Braun, G., Bevis, E. and Doncaster, K. (2006) A simple protocol for protein extraction of recalcitrant fruit tissues suitable for 2-DE and MS analysis. *Electrophoresis* 27: 3144–3151.

Sugiura, T., Kuroda, H., Honjo, H. and Ito, D. (2011) Temperature dependence of endodormancy development in flower buds of 'Kosui' Japanese pear and a model for estimating the completion of endodormancy. *Acta Hortic.* 587: 345–351.

Tamura, F., Tanabe, K. and Banno, K. (1992) Effect of chilling treatment on intensity of bud dormancy, respiration and endogenous growth regulators in Japanese pear 'Nijisseiki'. *J. Jpn. Soc. Hortic. Sci. (Japan)* 60: 763–769.

Tan, B.-C., Joseph, L.M., Deng, W.-T., Liu, L., Li, Q.-B., Cline, K., et al. (2003) Molecular characterization of the *Arabidopsis* 9-*cis* epoxycarotenoid dioxygenase gene family. *Plant J.* 35: 44–56.

Tuan, P.A., Bai, S., Saito, T., Imai, T., Ito, A. and Moriguchi, T. (2016) Involvement of *EARLY BUD-BREAK*, an AP2/ERF transcription factor gene, in bud break in Japanese pear (*Pyrus pyrifolia* Nakai) lateral flower buds: expression, histone modifications and possible target genes. *Plant Cell Physiol.* 57: 1038–1047.

Tuan, P.A., Bai, S., Yaegaki, H., Tamura, T., Hihara, S., Moriguchi, T., et al. (2015) The crucial role of *PpMYB10.1* in anthocyanin accumulation in peach and relationships between its allelic type and skin color phenotype. *BMC Plant Biol.* 15: 280.

Wan, C.Y. and Wilkins, T.A. (1994) A modified hot borate method significantly enhances the yield of high-quality RNA from cotton (*Gossypium hirsutum* L.). *Anal. Biochem.* 223: 7–12.

Wang, D., Du, P., Gao, Z., Xiao, W., Tan, Q., Chen, X., et al. (2016) Expression of ABA metabolism-related genes suggests similarities and differences between seed dormancy and bud dormancy of peach (*Prunus persica*). *Front. Plant Sci.* 6: 1248.

Welling, A. and Palva, E.T. (2008) Involvement of CBF transcription factors in winter hardiness in birch. *Plant Physiol.* 147: 1199–1211.

Wisniewski, M., Artlip, T. and Norelli, J. (2015) Overexpression of a peach CBF gene in apple: a model for understanding the integration of growth, dormancy, and cold hardiness in woody plants. *Front. Plant Sci.* 6: 85.

Wu, J., Seng, S., Sui, J., Vonapartis, E., Luo, X., Gong, B., et al. (2015) *Gladiolus hybridus* ABSCISIC ACID INSENSITIVE 5 (*Gh*ABI5) is an important transcription factor in ABA signaling that can enhance *Gladiolus* corm dormancy and *Arabidopsis* seed dormancy. *Front. Plant Sci.* 6: 960.

Wu, J., Wang, Z., Shi, Z., Zhang, S., Ming, R., Zhu, S., et al. (2013) The genome of the pear (*Pyrus bretschneideri* Rehd.). *Genome Res.* 23: 396–408.

Xiong, L. and Zhu, J.-K. (2003) Regulation of abscisic acid biosynthesis. *Plant Physiol.* 133: 29–36.

Yamane, H., Ooka, T., Jotatsu, H., Hosaka, Y., Sasaki, R. and Tao, R. (2011) Expressional regulation of *PpDAM5* and *PpDAM6*, peach (*Prunus persica*) dormancy-associated MADS-box genes, by low temperature and dormancy-breaking reagent treatment. *J. Exp. Bot.* 62: 3481–3488.

Yoshida, T., Fujita, Y., Maruyama, K., Mogami, J., Todaka, D., Shinozaki, K., et al. (2015) Four *Arabidopsis* AREB/ABF transcription factors function predominantly in gene expression downstream of SnRK2 kinases in abscisic acid signalling in response to osmotic stress. *Plant Cell Environ.* 38: 35–49.

Zheng, C., Halaly, T., Acheampong, A.K., Takebayashi, Y., Jikumaru, Y., Kamiya, Y., et al. (2015) Abscisic acid (ABA) regulates grape bud dormancy, and dormancy release stimuli may act through modification of ABA metabolism. *J. Exp. Bot.* 66: 1527–1542.

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